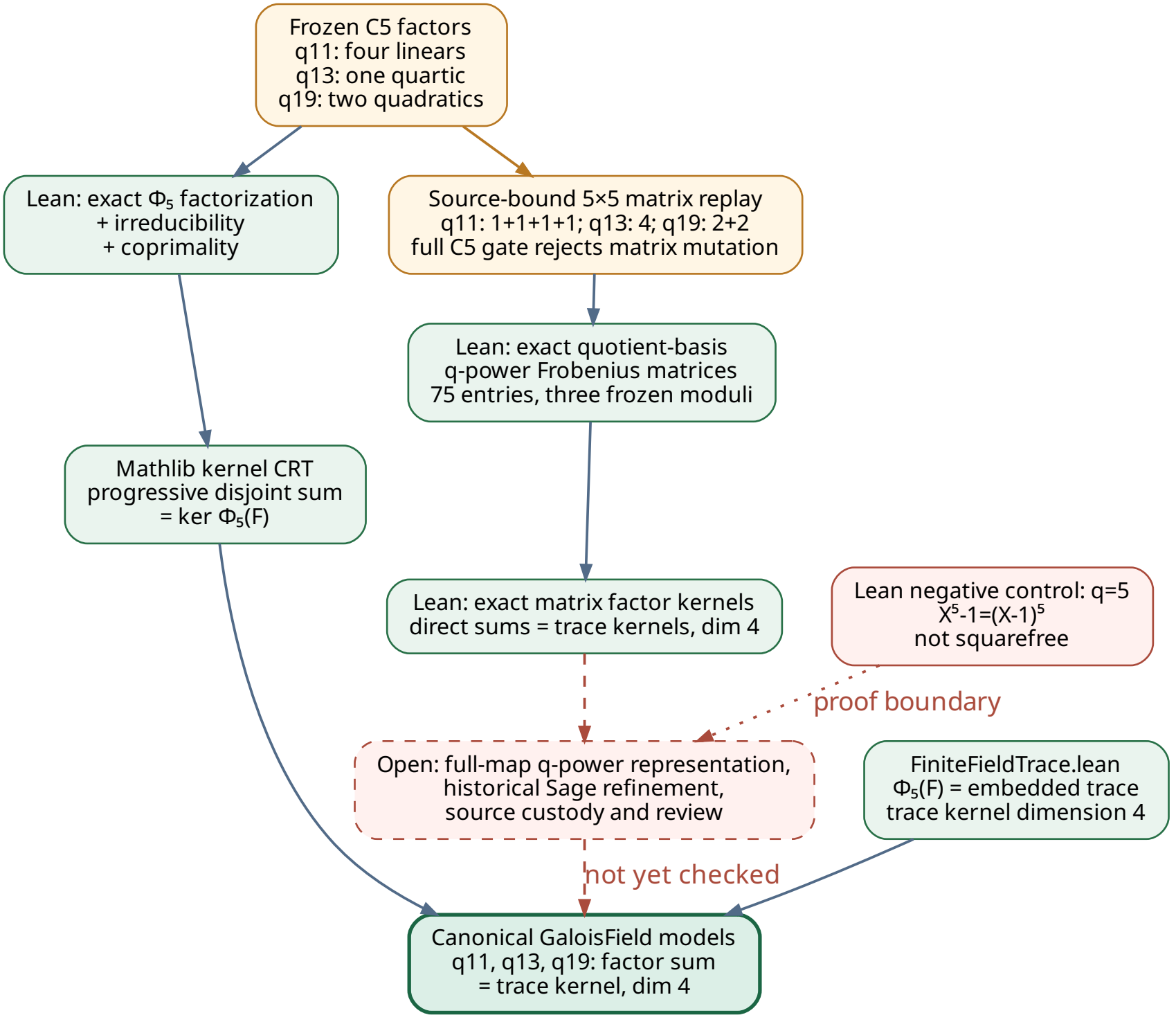
Frozen C5 factors
q11: four linears
q13: one quartic
q19: two quadratics
Lean: exact Φ_5 factorization
+ irreducibility
+ coprimality
Source-bound 5×5 matrix replay
q11: 1+1+1+1; q13: 4; q19: 2+2
full C5 gate rejects matrix mutation
Lean: exact quotient-basis
q-power Frobenius matrices
75 entries, three frozen moduli
Mathlib kernel CRT
progressive disjoint sum
= ker $\Phi_5(F)$
Lean: exact matrix factor kernels
direct sums = trace kernels, dim 4
Lean negative control: q=5
$X^5-1=(X-1)^5$
not squarefree
proof boundary
Open: full-map q-power representation,
historical Sage refinement,
source custody and review
FiniteFieldTrace.lean
$\Phi_5(F)$ = embedded trace
trace kernel dimension 4
not yet checked
Canonical GaloisField models
q11, q13, q19: factor sum
= trace kernel, dim 4